



BrainStation
1-800-903-5159

Unit 5: Professional Development

30%

Student ID

Student ID: 707576

Lakshmi Anathakrishnan

Area of Interest (5% of unit grade)						
			Meets Requirements	Partially Meets Requirements	Does Not Meet Requirements	Days Late
Categories	Problem Area & User	20%	X			
	Big Idea	20%	X			
	Impact	20%		X		
	Data	20%	X			
	Alternative	20%	X			

Grade: 90.0%

Capstone Sprint 1 (15% of unit grade)			Performance Rating						
			Exemplary	Very Good	Satisfactory	Developing	Limited	Incomplete	Days Late
Categories	GitHub	15%	X						
	Project Intro	25%	X						
	Data Acquisition & Exploration	25%	X						
	Communication & Visualization	25%		X					
	Presentation	10%			X				

Grade: 94.5%

Capstone Sprint 2 (30% of unit grade)			Performance Rating						
			Exemplary	Very Good	Satisfactory	Developing	Limited	Incomplete	Days Late
Categories	GitHub	5%	X						
	Data Processing	30%	X						
	EDA & Baseline Modeling	30%		X					
	Communication & Visualization	25%		X					
	Presentation	10%		X					

Grade: 93.5%

Capstone Sprint 3 (50% of unit grade)			Performance Rating						
			Exemplary	Very Good	Satisfactory	Developing	Limited	Incomplete	Days Late
Categories	GitHub	5%							
	Data Processing	10%							
	Modeling	50%							
	Communication & Visualization	20%							
	Presentation & Data Product	15%							

Grade:

Unit Grade

Unit Grade:

Additional Comments

Deliverable 1 - Areas of Interest

Great job on explaining both the problem area and the user! You also identify how machine learning can help with homeloan default prediction and give some examples of possible ML techniques you might employ throughout your project. Regarding the impact, while your answer outlines several potential benefits of home loan default prediction models, such as enhancing risk management practices, promoting fair lending practices, and fostering economic stability, it does so in qualitative terms. It would be nice to add a quantitative aspect. This could include the total value of the home loan market, how much is lost on defaults each year, and/or the rate of defaults. This will provide a rough sense of the value of this problem space. Lastly you have effectively identified and described multiple datasets relevant to the subject area, as well as a clear and concise summary of a problem in an alternative subject area.

Deliverable 2 - Sprint 1 Capstone

GitHub

Very engaging introduction, you have crafted a great README file.

Project Intro

- Well done on the project intro!
- Consider phrasing your areas of interest using a "how might we" statement. Examples can be found in the Capstone Introduction slides on Synapse.

Data Acquisition & Exploration

Very thoughtful approach to handling missing values, and very detailed data exploration steps. Good job!

Communication & Visualization

- Your visuals are excellent and effectively convey their meaning with labels and arranged values.
- However, for the pie charts and some bar graphs, titles are missing. Also, try to avoid overlapping numerical labels.

Presentation

You have chosen a good dataset which should lend itself to good business analysis. The slides were generally well organized, but could benefit from having less blocks of text. You often read directly from your slides which can undermine audience engagement. It would be nice to discuss more about the implications of dropping NULL values and the approach selected for filling NULL values. To further clarify and engage the audience, some visuals when discussing data exploration could be included as well as clearly state what the main findings were. The presentation ran a little long (8 minutes). Make sure to practice ahead of time and be clear on what your key points are.

Deliverable 3 - Sprint 2 Capstone

GitHub

- You've done an excellent job maintaining your GitHub repository, which accurately represents the current state of your project.
- The README file is well-organized and now effectively includes your EDA findings and baseline modeling efforts.

Data Processing

- Fantastic work on data processing! You've successfully tackled data quality issues, managed missing/duplicated values, and iterated on feature engineering.

EDA & Baseline Modeling

- Your thorough and detailed EDA has revealed significant insights into your data.
- Establishing baselines with logistic regression was a wise choice, providing interpretive and informative insights into your modeling.
- To further enhance your analysis and evaluation, consider utilizing evaluation models and metrics to assess and compare your results more comprehensively, and to understand the feature importance across different models.

Communication & Visualization

- Your communication is clear and well-structured, with appropriate visualizations.
- The notebook is well-organized, logically formatted, and presented in an easy-to-understand manner.
- Visualizations are clear, readable, and effectively highlight insights using suitable color combinations.

Presentation

You presented a clear slide stating the problem, solution, and impact, though it lacked quantitative estimates and focused on qualitative impact. Your explanation of the dataset and preprocessing was strong, but I suggest changing the plot as it was distracting and not very informative. The text, however, was great. Your key findings from exploratory data analysis (EDA) were succinct and effective. The baseline models' performance seemed a bit low, I would suggest reviewing it. In the advanced modeling and future work section, the plot was not easily perceptible, but your next steps were clear and logical. The use of SMOTE was well-justified and appropriate for your case.
